Module 5

Deep Learning
(Deep Neural Networks)

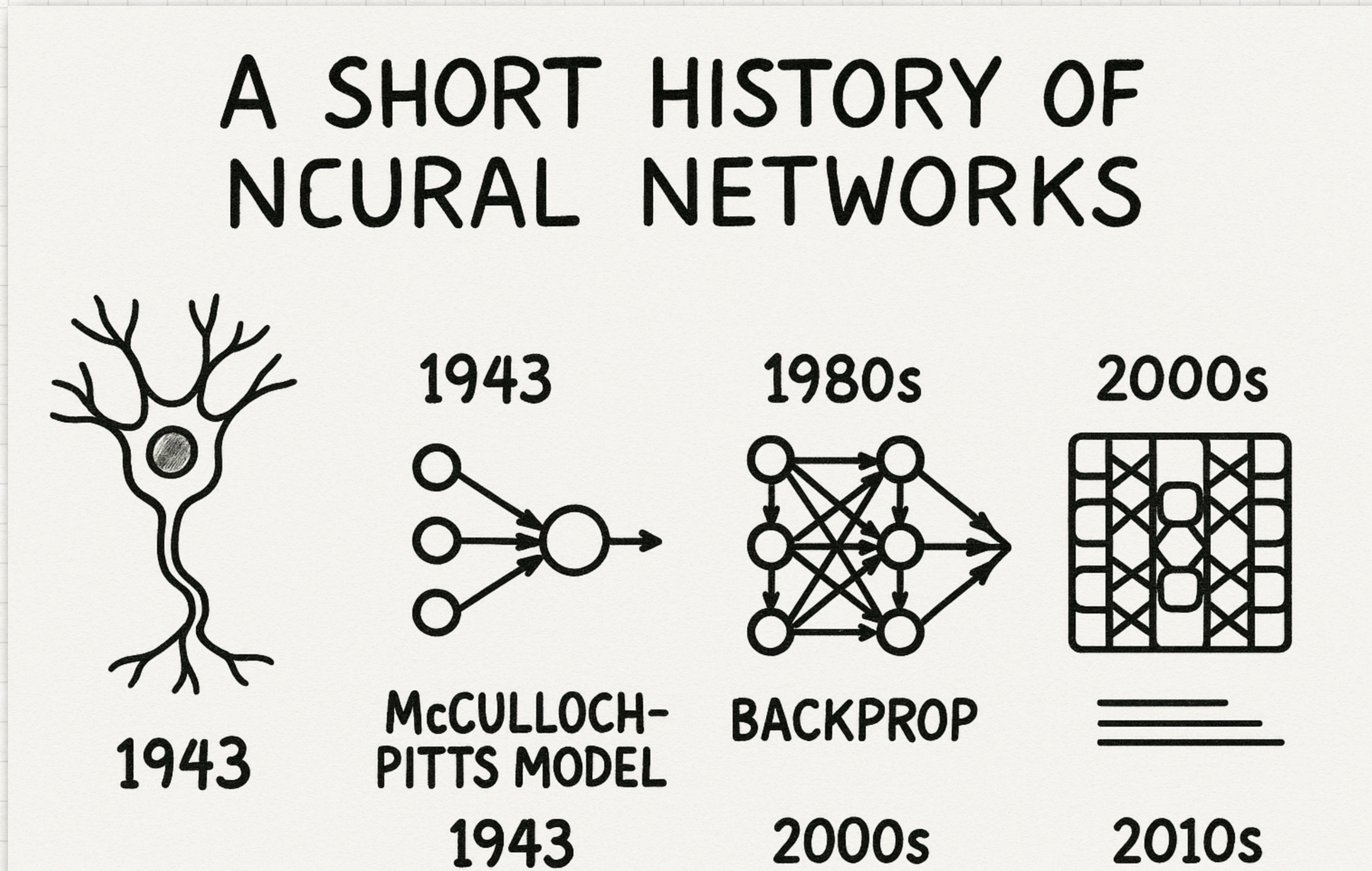
Resources

PyTorch a 60-minute blitz

Goodfellow et al. (2016) Deep Learning

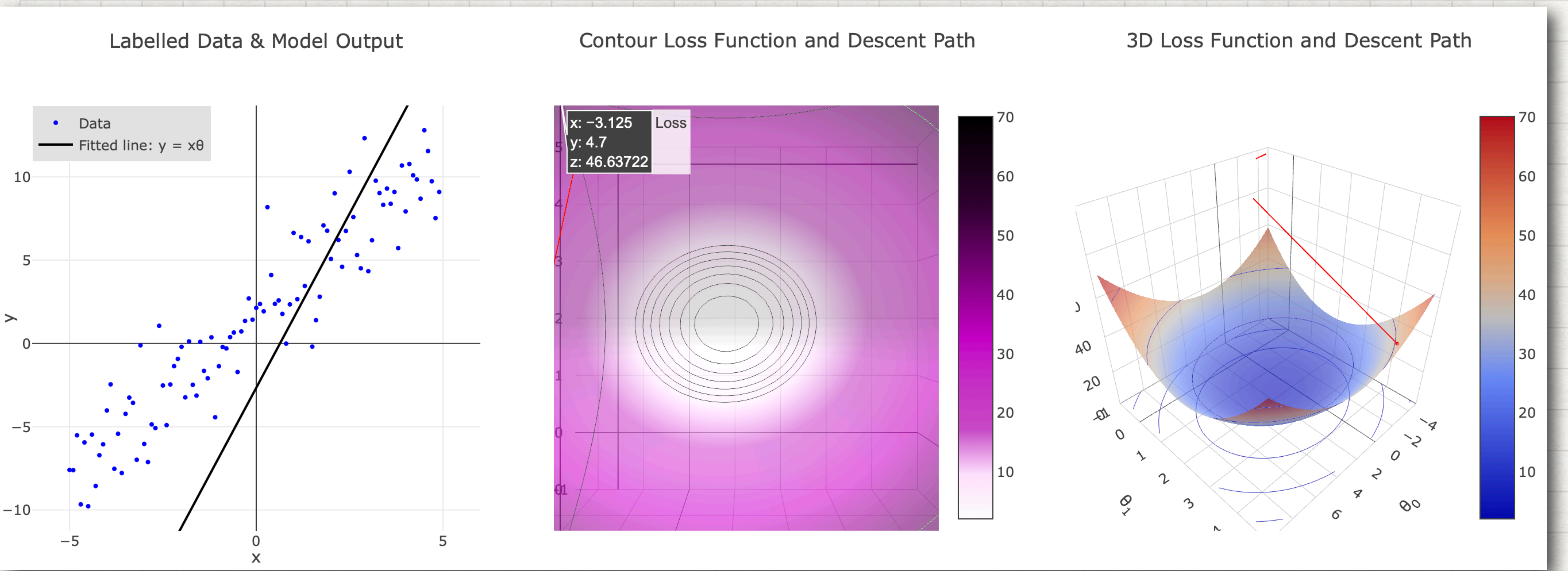
Deep Neural Networks

A very short history



Learning through back-propagation

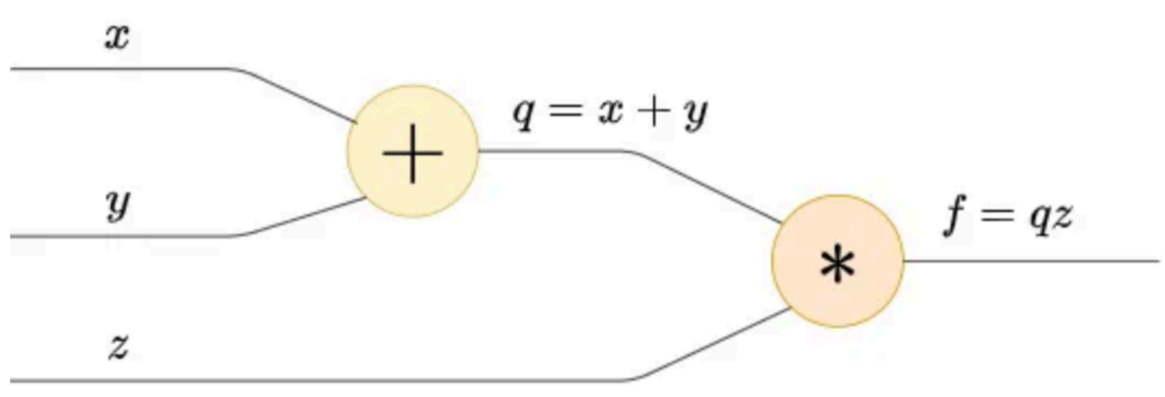
Gradient Descent



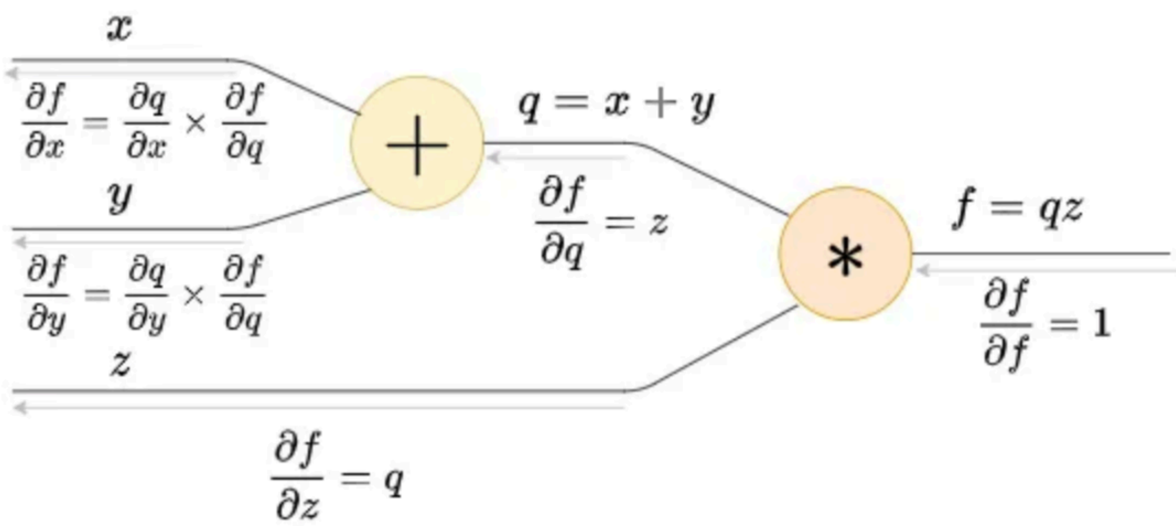
https://aero-learn.imperial.ac.uk/vis/Machine Learning/gradient_descent_3d.html

Learning through back-propagation

Credit assignment & backprop



forward pass



The light gray arrows represent backward pass

Support for Deep Learning in Python

PyTorch:

$ conda activate cog-ai

$ pip install torch torchvision

Tensors (Like numpy arrays, but better):

x = torch.tensor([2.0, 3.0])

Autograd:

x = torch.tensor([2.0, 3.0], requires_grad=True)

torch.nn:

A library containing different types of NN layers and operations on those layers:

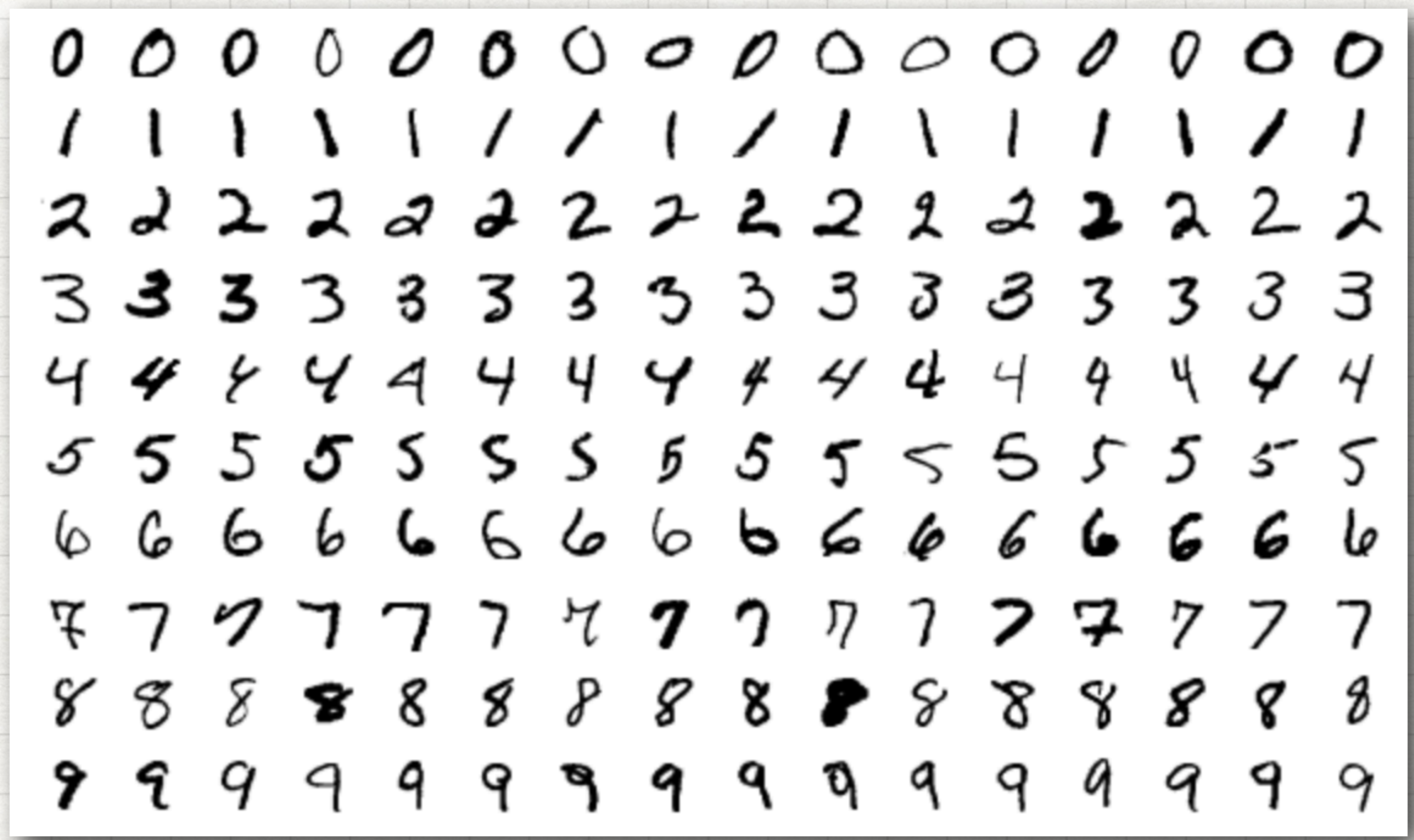
e.g. nn.Linear, nn.Sequential, nn.ReLU, nn.Conv2D, etc.

DataLoader:

An iterable utility that wraps a Dataset to provide an easy and efficient way to access data in batches. It handles tasks like automatically grouping samples into batches, shuffling the data each epoch, and parallelizing the data loading process across multiple worker processes

Mini-project 4: A small Neural Network

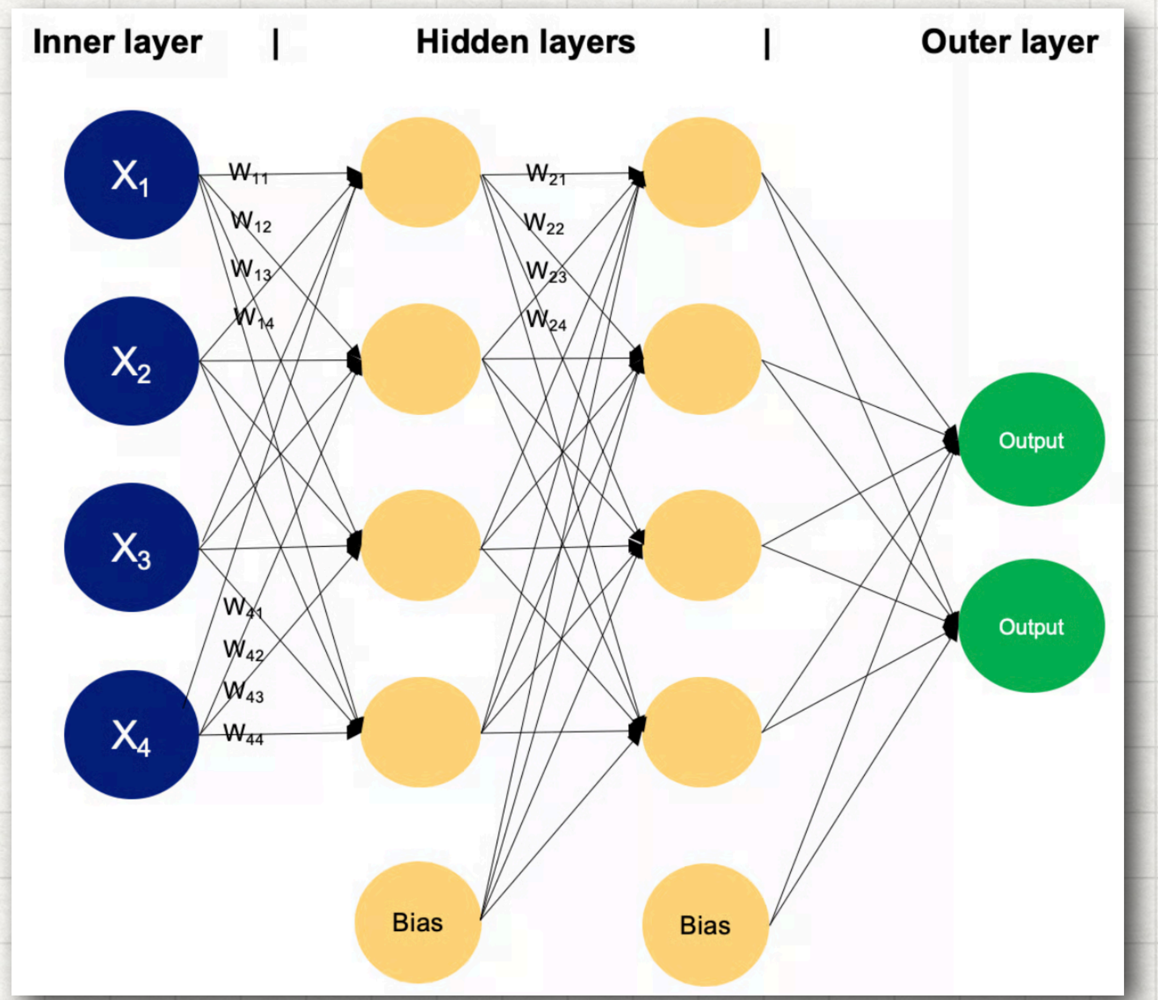
Train an MLP on MNIST and then analyse confusion patterns as a proxy for visual similarity.



MNIST Database

Mini-project 4: A small Neural Network

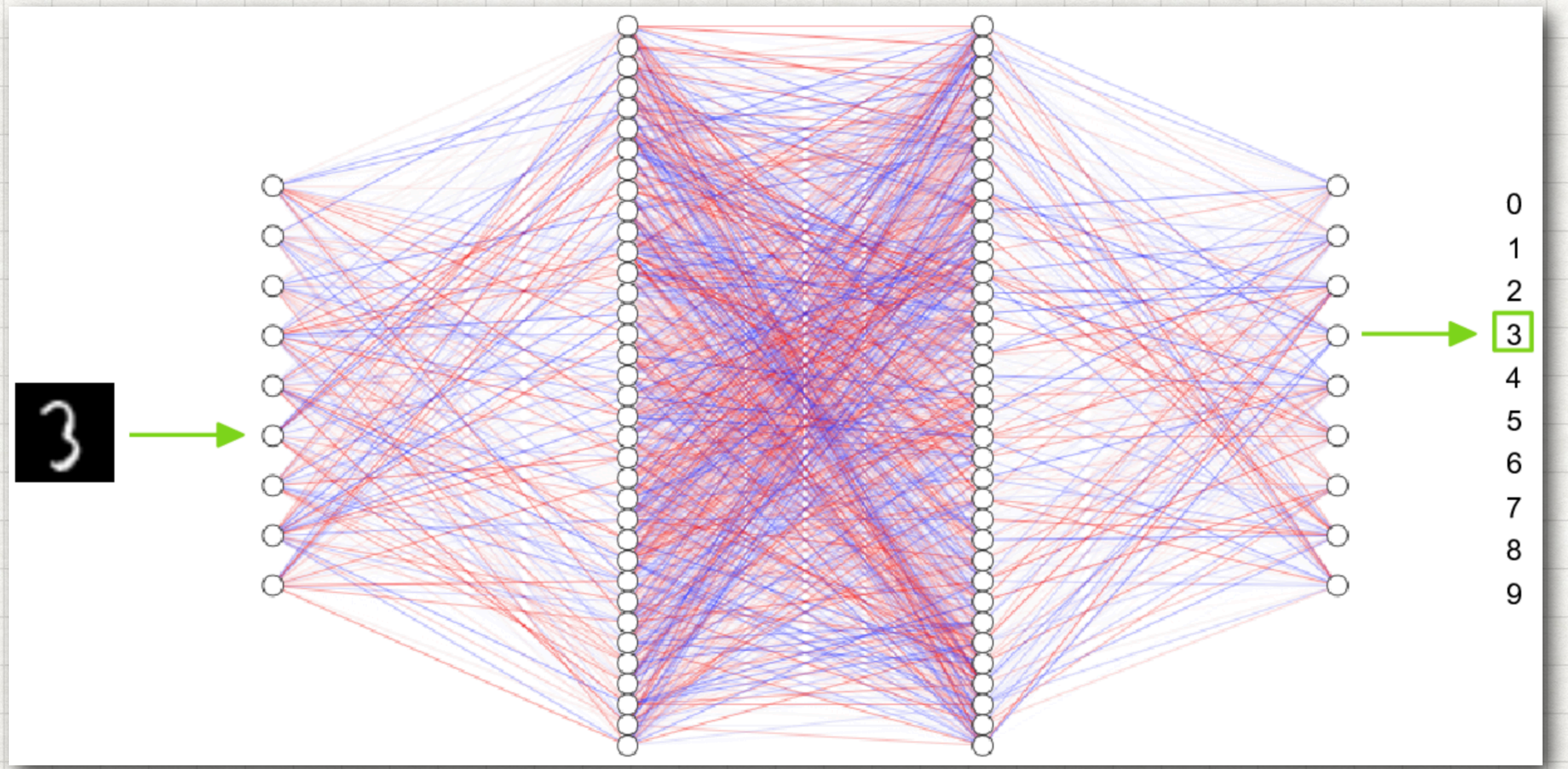
Train an MLP on MNIST and then analyse confusion patterns as a proxy for visual similarity.



Multi-layer Perceptron

Mini-project 4: A small Neural Network

Train an MLP on MNIST and then analyse confusion patterns as a proxy for visual similarity.



Demo 1: Using Tensors & Autograd

Define tensors 'a' and 'b'. Perform series of calculations on these tensors, using variables 'c', 'd', and 'e' to store outputs. Then print the gradient of 'e' with respect to 'a' and 'b'.

$$c = a + b$$
$$d = b + 1$$
$$e = c * d$$

Compute: $\dfrac{\delta e}{\delta a} \quad \& \quad \dfrac{\delta e}{\delta b}$

Demo 1: Using Tensors & Autograd

Computational Graph